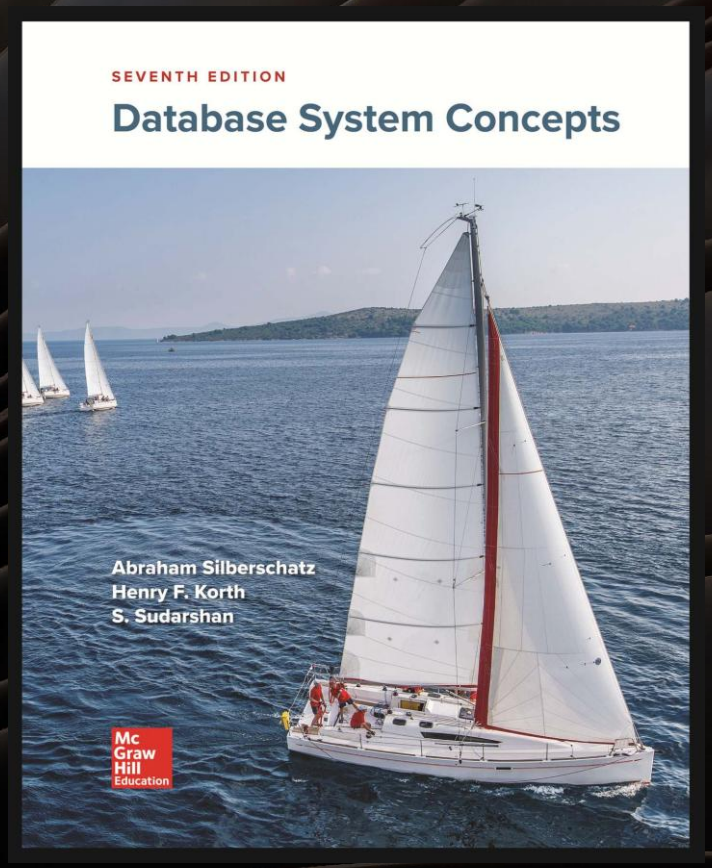


IF2040 – Pemodelan Basis Data

Relational Algebra

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Sumber

Silberschatz, Korth, Sudarshan: "Database System Concepts", 7th Edition

- Chapter 2: Relational Model
- Section 2.5. Relational Query Languages
- Section 2.6. The Relational Algebra
- Page 47 - 58

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Relational Algebra

Procedural language

Six basic operators

- select: σ
- project: Π
- union: $\cup$
- set difference: $-$
- cartessian product: $\times$
- rename: ρ

The operators take one or two relations as inputs and produce a new relation as a result.

- Additional Operators
  - Intersection: $\cap$
  - natural join: $\bowtie$
  - assignment: $\leftarrow$
  - outer join: $⟗$
  - division: $\div$
- Extended Operators
  - generalized projection
  - aggregation



Additional Operators

Set-Intersection Operation

Notation

$$r \cap s$$

Defined as

$$r \cap s = \{ t \mid t \in r \text{ and } t \in s \}$$

Requirement

Set intersection must be taken between *compatible* relations.

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Set-Intersection Operation – Example 1

RELATIONS

R

A	B
α	1
α	2
β	1

S

A	B
α	2
β	3

R ∩ S:

A	B
α	2

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Set-Intersection Operation

Define set-intersection by using basic operators

$$r \cap s = r - (r - s)$$

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Natural-Join Operation

Notation

$$r \bowtie s$$

Defined as

- Let r and s be relations on schemas R and S respectively.
  Then, $r \bowtie s$ is a relation on schema $R \cup S$ obtained as follows:
  - If t_r and t_s have the same value on each of the attributes in $R \cap S$, add a tuple t to the result (concatenation of t_r and t_s)

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Natural Join Operation – Example 1

RELATIONS

R

A	B	C	D
α	1	α	a
β	2	γ	a
γ	4	β	b
α	1	γ	a
δ	2	β	b

S

B	D	E
1	a	α
3	a	β
1	a	γ
2	b	δ
3	b	ε

R ⋈ S

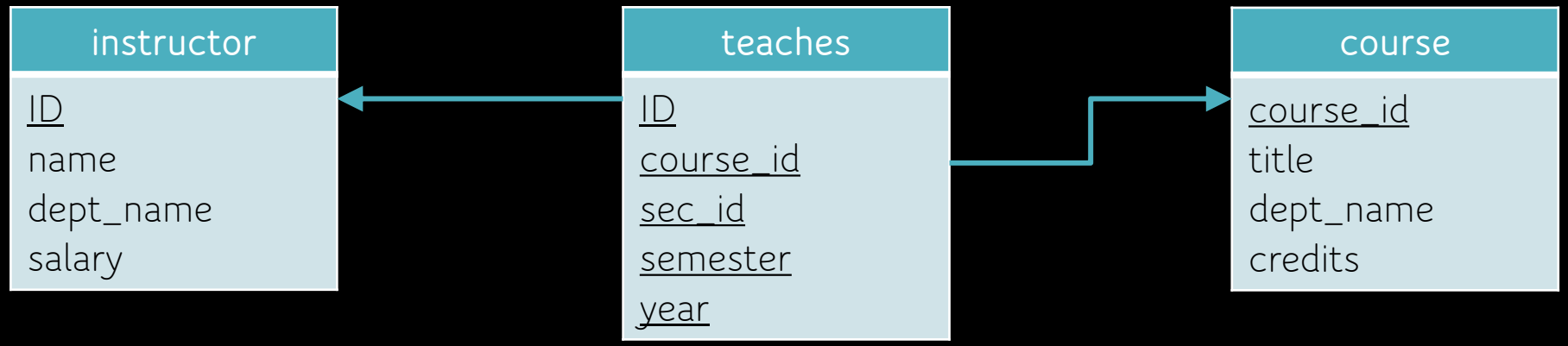
A	B	C	D	E
α	1	α	a	α
α	1	α	a	γ
α	1	γ	a	α
α	1	γ	a	γ
δ	2	β	b	δ

Note: natural join can be expressed using basic operations.
e.g. $r \bowtie s$ can be written as

$$\Pi_{r.A,\ r.B,\ r.C,\ r.D,\ s.E}\ (\sigma_{r.B = s.B \wedge r.D = s.D}\ (r \times s))$$

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Natural Join Operation – Example 2



Find the names of all instructors in the Comp. Sci. department together with the course titles of all the courses that the instructors teach

$\Pi_{name,\ title}(\sigma_{dept_name=\text{"Comp. Sci."}}(instructor \bowtie teaches \bowtie course))$

$\Pi_{name,\ title}(\sigma_{dept_name=\text{"Comp. Sci."}}(instructor \bowtie teaches \bowtie \Pi_{course_id,\ title}(course)))$

dept_name should not be used as joined attribute

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Natural Join and Theta Join

Natural join is associative

$(instructor \bowtie teaches) \bowtie course$ is equivalent to
$instructor \bowtie (teaches \bowtie course)$

Natural join is commutative

$instructor \bowtie teaches$ is equivalent to
$teaches \bowtie instructor$

The **theta join** operation

$r \bowtie_\theta s = \sigma_\theta (r \times s)$

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Assignment Operation

The assignment operation (←) provides a convenient way to express complex queries.

- Write query as a sequential program consisting of
  - a series of assignments
  - followed by an expression whose value is displayed as a result of the query.
- Assignment must always be made to a temporary relation variable.

Example: Find all instructor in the "Physics" and "Music" department.

$Physics \leftarrow \sigma_{dept_name="Physics"}(instructor)$

$Music \leftarrow \sigma_{dept_name="Music"}(instructor)$

$Physics \cup Music$

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Outer Join

Notation:

- Left outer join: ⟕
- Right outer join: ⟖
- Full outer join: ⟗

Defined as

- Computes the join and then adds tuples from one relation that does not match tuples in the other relation to the result of the join.

Uses *null* values

- *null* signifies that the value is unknown or does not exist
- All comparisons involving *null* are (roughly speaking) **false** by definition.

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Outer Join Examples

Relation *course*

course_id	*title*	*dept_name*	*credits*
BIO-301	Genetics	Biology	4
CS-190	Game Design	Comp. Sci.	4
CS-315	Robotics	Comp. Sci.	3

Relation *prereq*

course_id	*prereq_id*
BIO-301	BIO-101
CS-190	CS-101
CS-347	CS-101

Observe that

course information is missing CS-347

prereq information is missing CS-315

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Left Outer Join

Command:

course ⟕ *prereq*

course_id	*title*	*dept_name*	*credits*	*prereq_id*
BIO-301	Genetics	Biology	4	BIO-101
CS-190	Game Design	Comp. Sci.	4	CS-101
CS-315	Robotics	Comp. Sci.	3	*null*

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Right Outer Join

Command:

course ⟖ *prereq*

course_id	*title*	*dept_name*	*credits*	*prereq_id*
BIO-301	Genetics	Biology	4	BIO-101
CS-190	Game Design	Comp. Sci.	4	CS-101
CS-347	*null*	*null*	*null*	CS-101

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Full Outer Join

Command:

course ⟗ *prereq*

course_id	*title*	*dept_name*	*credits*	*prereq_id*
BIO-301	Genetics	Biology	4	BIO-101
CS-190	Game Design	Comp. Sci.	4	CS-101
CS-315	Robotics	Comp. Sci.	3	*null*
CS-347	*null*	*null*	*null*	CS-101

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Outer Join Operation

Express outer join (e.g. r ⟗ s) using basic [and join – to simplify] operations

$$(r \bowtie s) \cup ((r - \Pi_R (r \bowtie s)) \times \{(null, \ldots, null)\})$$

Constant relation with only 1 tuple

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Division Operation

Notation

$$r \div s$$

Defined as

Given relations r(R) and s(S), such that $S \subseteq R$,
$r \div s$ is the largest relation t(R-S) such that $t \times s \subseteq r$

E.g.

let $r(ID, course_id) = \prod_{ID,\ course_id}(takes)$ and
$s(course_id) = \prod_{course_id}(\sigma_{dept_name="Biology"}(course))$
then $r \div s$ = students who have taken all courses in the Biology Dept.

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Division Operation

r

A	B
α	1
α	2
α	3
β	1
γ	1
δ	1
δ	3
δ	4
ε	6
ε	1
β	2

s

B
1
2

$r \div s$

A
α
β

Express r $\div s$ using basic operations

$$temp1 \leftarrow \prod_{R-S} (r)$$
$$temp2 \leftarrow \prod_{R-S} ((temp1 \times s) - \prod_{R-S,S} (r))$$
$$result = temp1 - temp2$$

In relation R, α and β are paired with all values of B in relation S,
i.e. tuples (α,1), (α,2), (β,1), (β,2) exist

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